

Science — Cbse Half-Yearly Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	Assertion (A): Uncontrolled cell divisions leading to tumour formation are a result of errors in mitosis, rather than errors in meiosis. Reason (R): Faulty meiosis may cause early pregnancy loss or reduced fertility, while errors in mitosis lead to an abnormal number of chromosomes in body cells. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. Both the assertion and the reason are true statements derived from the consequences of errors in cell division. The reason does not correctly explain the assertion because listing the separate outcomes of faulty meiosis and mitosis errors does not explain why uncontrolled divisions specifically result from mitosis errors. [Ch 2: Cell: The Building Block of Life]	1
2	What is the key difference between scalar and vector quantities? (A) Scalars are specified by just their numerical value, while vectors require specifying both direction and magnitude. (B) Scalars have both magnitude and direction, while vectors have only magnitude. (C) Scalars can only be positive, while vectors can be positive or negative. (D) Scalars are used for time, while vectors are used for distance. Answer: Correct Option: (A) ► Why? A. Scalars can be specified by just their numerical value, while vectors require specifying both direction and magnitude.	1



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 4: Describing Motion Around Us]</i>	
3	When studying the motion of a falling object, what assumption is made in building a model? (A) The object's shape is perfectly spherical (B) The object falls at a constant speed (C) Gravity varies depending on the object's mass (D) Air resistance is neglected to understand the basic effect of gravity Answer: Correct Option: (D) ► Why? D. When studying the motion of a falling object, air resistance may be neglected to understand the basic effect of gravity. This is a deliberate simplification to focus on what is most important. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1
4	On a graph paper, the scale chosen for time on the x-axis is 5 divisions = 1 s. How many divisions will represent 4 s on the x-axis? (A) 9 divisions (B) 10 divisions (C) 20 divisions (D) 15 divisions Answer: Correct Option: (C) ► Why? C. If 5 divisions represent 1 s, then 4 s is represented by $4 \times 5 = 20$ divisions. <i>[Ch 4: Describing Motion Around Us]</i>	1
5	Which tissue is responsible for the increase in length and girth of plants? (A) Connective tissue (B) Skeletal tissue (C) Meristematic tissue (D) Nervous tissue Answer: Correct Option: (C) ► Why? C. Growth in plants requires actively dividing cells that together form a tissue called a meristematic tissue. <i>[Ch 3: Tissues in Action]</i>	1



Q.NO.	QUESTIONS	MARKS
6	Assertion (A): In an oxygen molecule, the two oxygen atoms are joined by a double bond. Reason (R): Two oxygen atoms share two electrons each to complete their octet, forming two pairs of shared electrons. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? A. The assertion is true because an oxygen molecule involves two pairs of shared electrons, constituting a double bond. The reason correctly explains this by stating that two oxygen atoms share two electrons each, forming two pairs of shared electrons to complete their octet. [Ch 9: Atomic Foundations of Matter]	1
7	In the airplane fuel incident, approximately how many litres of fuel was the aircraft short? (A) About 10,000 litres (B) About 15,000 litres (C) About 20,000 litres (D) About 22,300 litres Answer: Correct Option: (B) ► Why? B. The aircraft was about 15,000 litres short of fuel due to the unit mix-up in calculating fuel density. [Ch 1: Exploration: Entering the World of Secondary Science]	1
8	When a coconut is brought down at a very high velocity to hit a hard surface and stops in a very short time, what causes the shell to break? (A) A decrease in the mass of the coconut (B) A small force applied over a long time (C) A very large force exerted by the ground to change its velocity quickly (D) The gravitational force of the Earth pulling it downwards Answer: Correct Option: (C) ► Why? C. To change the coconut's velocity so quickly (in a very short time), the ground must exert a very large force on it, which breaks the shell. [Ch 6: How Forces Affect Motion]	1



Q.NO.	QUESTIONS	MARKS
9	Assertion (A): Whenever one object exerts a force on a second object, the second object simultaneously exerts an equal and opposite force on the first object. Reason (R): Newton's third law of motion applies to all types of forces, whether contact or non-contact. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. The assertion is true because it is the direct statement of Newton's third law of motion. The reason is also true but does not explain the assertion, as it only describes the scope of the law's applicability rather than defining the law itself. [Ch 6: How Forces Affect Motion]	1
10	In the process of coagulation, what does a coagulant do? (A) Makes smaller particles clump together and settle down (B) Dissolves the solute completely in the solvent (C) Spins the mixture rapidly to separate components (D) Converts a liquid directly into vapour Answer: Correct Option: (A) ► Why? A. Coagulation involves the addition of a coagulant that makes smaller particles clump together and settle down in a heterogeneous mixture of solids and liquids. [Ch 5: Exploring Mixtures and their Separation]	1
11	Mature Red Blood Cells (RBCs) in humans have a short lifespan because they lack a nucleus and cannot repair or divide. Approximately how long do they survive? (A) 7 days (B) 30 days (C) 120 days (D) 365 days Answer: Correct Option: (C) ► Why? C. Because mature RBCs lack a nucleus and cannot repair or divide themselves, their lifespan is short and they survive for approximately 120 days. [Ch 2: Cell: The Building Block of Life]	1
12	Raja Ramanna is often called the Father of the Indian Nuclear Programme. What did he promote? (A) The use of nuclear weapons for defence (B) The peaceful use of nuclear energy for development (C) The replacement of all power plants with nuclear plants (D) The export of nuclear technology to other nations	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (B) ► Why? B. Raja Ramanna made significant contributions in developing the nation's Nuclear Energy Programme and promoted its peaceful use for development. [Ch 9: Atomic Foundations of Matter]	
13	In which direction are planetary winds deflected in the Northern Hemisphere due to the Earth's rotation? (A) Towards the right (B) Towards the left (C) Upwards into the atmosphere (D) Downwards towards the surface Answer: Correct Option: (A) ► Why? A. The Earth's rotation causes planetary winds to be deflected towards the right in the Northern Hemisphere and towards the left in the Southern Hemisphere. [Ch 13: Earth as a System: Energy, Matter, and Life]	1
14	When an external force is removed from a deformed spring, the spring returns to its original shape. What causes this return to the original shape? (A) Internal forces undoing the deformation (B) External gravitational force (C) The change in the mass of the spring (D) The kinetic energy of the spring Answer: Correct Option: (A) ► Why? A. Once the external force is removed, the internal forces in the spring undo the deformation. [Ch 7: Work, Energy, and Simple Machines]	1
15	Assertion (A): Tendons connect bone to bone and allow joint movement. Reason (R): Tendons are made of tough connective tissue that transmits force from muscle to bone. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? D. The assertion is false because tendons connect muscle to bone to transmit force, whereas ligaments connect bone to bone. The reason is true as it accurately describes the composition and function of tendons. [Ch 3: Tissues in Action]	1



Q.NO.	QUESTIONS	MARKS
16	In 1958, F.C. Steward demonstrated that cells from the vascular phloem of carrot can regenerate whole plants. What specific components were present in the nutrient medium used in this experiment? (A) Simple sugars and hormones (B) Complex sugars and minerals (C) Simple sugars and antibiotics (D) Complex sugars and hormones Answer: Correct Option: (A) ► Why? A. F.C. Steward grew the phloem cells in a nutrient medium containing simple sugars and hormones under appropriate conditions to demonstrate totipotency. <i>[Ch 3: Tissues in Action]</i>	1
17	Why do plants look wilted when they do not get enough water? (A) The lysosomes break down the cell wall. (B) The Golgi apparatus stops packaging proteins. (C) The mitochondria halt energy production. (D) The vacuole loses water, cells become less firm, and the plant wilts. Answer: Correct Option: (D) ► Why? D. When a plant does not get enough water, the vacuole loses water, causing the cells to become less firm and the plant to wilt. <i>[Ch 2: Cell: The Building Block of Life]</i>	1
18	Assertion (A): An equation in science is primarily a calculation tool designed for finding numerical answers. Reason (R): Mathematics in science is a powerful language for thinking that helps express compact statements about how quantities are related. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? The assertion is false because an equation in science is explicitly described as not just a calculation tool, but a compact statement about how things are related. The reason is true because mathematics is described as a language for thinking and expressing relationships, not merely for finding numerical answers. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1



Q.NO.	QUESTIONS	MARKS
19	Assertion (A): We cannot hear the sound of a bell ringing in a closed jar after most of the air is pumped out. Reason (R): Sound requires a medium to travel. (i) Both A and R are true, but R is not the correct explanation of A. (ii) Both A and R are true, and R is the correct explanation of A. (iii) A is true, but R is false. (iv) A is false, but R is true. Answer: Correct Option: ((ii)) ► Why? Option (ii) is correct because both the Assertion and Reason are true, and the Reason correctly explains the Assertion. As demonstrated by the vacuum bell jar experiment, when air is pumped out of the jar, the sound becomes fainter and almost no sound can be heard even though the bell is ringing. This happens because sound needs a medium to propagate and cannot travel in a vacuum. Therefore, the reason we cannot hear the bell in the closed jar after the air is removed is precisely because the medium required for sound to travel has been taken away. [Ch 10: Sound Waves: Characteristics and Applications]	1
20	The biological sex of a baby is determined by the chromosome contributed by: (A) The mother only (B) Both parents equally (C) The father, who contributes either X or Y (D) The mother's hormone levels Answer: Correct Option: (C) ► Why? C. The mother always contributes an X chromosome, while the father contributes either an X (resulting in XX, female) or a Y (resulting in XY, male), thereby determining the baby's biological sex. [Ch 11: Reproduction: How Life Continues]	1
Section B Attempt 5 out of 7.		
21	At what minimum difference in boiling points can two miscible liquids be separated by distillation? Answer: Two miscible liquids can be separated by distillation if their boiling points differ by at least about 25 °C. A large difference in boiling points allows one liquid to vaporise before the vapours of the other liquid are formed in significant amount, enabling effective separation. ► Why? The source specifies that distillation works for liquids differing in boiling point by at least about 25 °C, as this ensures one liquid vaporises predominantly before the other. [Ch 5: Exploring Mixtures and their Separation]	2
	OR What is the difference between a law and a theory in science?	



Q.NO.	QUESTIONS	MARKS
	Answer: A law usually describes a regular pattern observed in nature, often expressed using words or mathematical relationships. A theory goes further and provides an explanation of why those patterns occur, usually based on evidence gathered over time. ► Why? This tests the distinction between two key scientific terms as explicitly defined in the source, requiring understanding of their different roles. [Ch 1: Exploration: Entering the World of Secondary Science]	
22	Explain why the cell membrane model is called the 'fluid mosaic' model. Answer: The model is called 'fluid' because the molecules can move sideways, flip, and rotate within the membrane. It is called 'mosaic' because the molecules are arranged like tiles in a mosaic, with proteins embedded in a lipid bilayer. ► Why? The source describes 'fluid' referring to molecular movement and 'mosaic' referring to the tile-like arrangement of molecules, justifying the model's name. [Ch 2: Cell: The Building Block of Life]	2
	OR An object of mass 5 kg is near the Earth's surface. Calculate the gravitational force acting on it. (Take $g = 9.8 \text{ m s}^{-2}$) Answer: Using $F = mg$, the gravitational force is $F = 5 \text{ kg} \times 9.8 \text{ m s}^{-2} = 49 \text{ N}$. This force acts in the downward direction towards the Earth. ► Why? The formula $F = mg$ from the source is applied with the given mass and the value of $g = 9.8 \text{ m s}^{-2}$ stated in the source. [Ch 6: How Forces Affect Motion]	
23	Define 'rate of change' and state how it relates to average velocity. Answer: The rate of change is the ratio of change in one quantity to the corresponding change in time. Average velocity is the average rate of change of position with respect to time, calculated as displacement divided by the time interval. ► Why? Tests understanding of the general concept of rate of change and its specific application to average velocity. [Ch 4: Describing Motion Around Us]	2
24	What compounds make the bone matrix rigid, and what is the nature of the cartilage matrix? Answer: The bone matrix contains calcium and phosphorus compounds, giving it strength and rigidity. In contrast, the cartilage matrix is soft and jelly-like, providing flexibility and cushioning. ► Why? The answer requires recalling the specific minerals in bone matrix and contrasting it with the qualitative nature of the cartilage matrix described in the source.	2



Q.NO.	QUESTIONS	MARKS
	[Ch 3: Tissues in Action]	
25	Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg^{2+}. However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride. Answer: Cl^-, one, two ► Why? According to the textbook context, a chlorine atom gains one electron to acquire a negative charge, becoming a chloride anion represented as Cl^-. Since a magnesium atom gives two electrons to become Mg^{2+}, and each chlorine atom can only accept one electron, it requires two chlorine atoms to accept the two electrons. Therefore, one ion of magnesium and two ions of chlorine combine to form magnesium chloride, ensuring that the positive and negative charges balance each other to create a neutral overall structure. ► Step-by-step: - Step 1: Identify the ion formed by chlorine. The context states chlorine gains one electron to become a chloride anion, Cl^-. - Step 2: Determine the ratio of ions. Magnesium forms a Mg^{2+} ion by giving two electrons. Since each Cl atom takes only one electron, two Cl^- ions are needed to accept the two electrons from one Mg^{2+} ion. - Step 3: Fill in the blanks with 'Cl^-', 'one', and 'two'. [Ch 9: Atomic Foundations of Matter]	2
Section C Attempt 5 out of 7.		
26	When creating a simple model of a cricket ball being hit for a six, on what basis do scientists decide which details to include and which to ignore? Give one example of each. Answer: In scientific modeling, details are included or ignored based on their impact on the outcome being studied. Details that make a significant difference to the result are kept, while those that make no difference are ignored to simplify the model. For example, when modelling a cricket shot to see if the ball crosses the boundary, the mass of the ball and the speed and direction of the hit are included because they are very important. Conversely, details like the brand of the bat or the colour of the ball are ignored because they make no difference to whether the ball crosses the boundary. ► Why? The question tests the student's understanding of scientific simplification and modeling, specifically how scientists prioritize relevant variables over irrelevant ones based on impact. ► Step-by-step: - Step 1: Identify the core principle of simplification, which is to include details that make a significant difference and ignore those that do not. - Step 2: Provide an example of details to include (mass, speed, direction) because they directly affect the ball's trajectory. - Step 3: Provide an example of details to ignore (bat brand, ball colour) because they have no effect on the outcome in a simple model.	3



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	
	OR Define the momentum of an object. How does Newton's second law of motion apply to situations where the mass of the object is not constant? Answer: The momentum of an object is defined as the product of its mass and velocity. The direction of the momentum is the same as that of the velocity. According to Newton's second law of motion, the rate of change of momentum of an object is proportional to the net force and takes place in the direction in which the net force acts. This expression of Newton's second law is more complete and is applicable to situations even where the mass of the object is not constant, unlike the $F = ma$ form which assumes constant mass. ► Why? Momentum defines motion by combining mass and velocity, and Newton's second law in terms of momentum's rate of change applies universally, even with changing mass. ► Step-by-step: 1. Step 1: Define momentum as the product of mass and velocity, with direction same as velocity 2. Step 2: State Newton's second law in terms of the rate of change of momentum 3. Step 3: Conclude that this form is applicable even when the mass of the object is not constant <i>[Ch 6: How Forces Affect Motion]</i>	
27	When cooking oil is mixed with water, it forms a separate layer. Which substance will form the top layer and which apparatus is used to separate these immiscible liquids? On what property is this separation based? Answer: When cooking oil is mixed with water, they form two separate layers because they are immiscible liquids. Since oil is less dense than water, the cooking oil will form the top layer, while the denser water will settle at the bottom. The apparatus used to separate these two immiscible liquids is a separating funnel. This device allows the denser liquid (water) to be drained off through a stopcock at the bottom, leaving the lighter liquid (oil) behind. The separation is based on the different densities of the two immiscible liquids, which causes them to form distinct layers that can be easily decanted. ► Why? Oil floats on water due to its lower density. A separating funnel exploits the density difference to separate immiscible liquids. ► Step-by-step: 1. Step 1: Identify that cooking oil is less dense than water, so it forms the top layer. 2. Step 2: Name the apparatus used for separating immiscible liquids as the separating funnel. 3. Step 3: State that the separation is based on the different densities of the two immiscible liquids. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	3
	OR Define average acceleration. How does its direction relate to the velocity when the magnitude of velocity is decreasing in a straight line motion?	



Q.NO.	QUESTIONS	MARKS
	Answer: Average acceleration is defined as the change in velocity divided by the time interval over which the change occurs. The formula is: average acceleration = (final velocity - initial velocity) / time interval. Like displacement and velocity, average acceleration requires specifying both magnitude and direction. For motion in a straight line, if the magnitude of velocity is decreasing, the average acceleration acts opposite to the direction of velocity. Conversely, if the magnitude of velocity is increasing, the acceleration is in the direction of velocity. ► Why? Average acceleration direction matches the change in velocity. If speed decreases, the acceleration opposes the current velocity direction. ► Step-by-step: 1. Step 1: Define average acceleration as change in velocity divided by time interval. 2. Step 2: Note that acceleration has both magnitude and direction. 3. Step 3: State that if velocity magnitude is decreasing, acceleration is opposite to the direction of velocity. [Ch 4: Describing Motion Around Us]	
28	Differentiate between Rough Endoplasmic Reticulum (RER) and Smooth Endoplasmic Reticulum (SER) based on their structure and primary functions. Answer: The Endoplasmic Reticulum (ER) has two distinct types based on its structure and function. Rough Endoplasmic Reticulum (RER) looks rough under an electron microscope because it has ribosomes attached to its surface. It is mainly involved in protein synthesis and protein secretion, such as in gland cells like pancreatic cells. In contrast, Smooth Endoplasmic Reticulum (SER) does not have ribosomes on its surface, making it look smooth. Its primary function is the synthesis and storage of fats (lipids) and hormones within the cell. ► Why? RER and SER are distinguished by the presence or absence of ribosomes, which directly dictates their visual appearance and their specialized roles in synthesizing proteins versus fats and hormones. ► Step-by-step: 1. Step 1: Note the structural difference: RER has ribosomes (looks rough), while SER lacks ribosomes (looks smooth). 2. Step 2: Identify RER's function: protein synthesis and secretion (e.g., in pancreatic cells). 3. Step 3: Identify SER's function: synthesis and storage of fats and hormones. [Ch 2: Cell: The Building Block of Life]	3
29	How does a movable pulley system differ from a fixed pulley in terms of mechanical advantage? Give one real-life application of pulley systems.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Unlike a fixed pulley, a movable pulley or a system of pulleys can have a mechanical advantage greater than 1. In a movable pulley system, the load is attached to the movable pulley itself, with one end of the rope fixed to a point and the other end free to apply effort. Because the mechanical advantage is greater than 1, these systems allow us to lift much heavier objects with a much smaller effort. Pulley systems are widely used in real life, such as in elevators and cranes, where they provide the convenience of lifting heavy loads vertically with reduced effort. ► Why? Movable pulleys have a mechanical advantage greater than 1, allowing heavy loads to be lifted with less effort, unlike fixed pulleys which only change direction. ► Step-by-step: 1. Step 1: Contrast movable pulleys with fixed pulleys by stating movable pulleys have a mechanical advantage greater than 1. 2. Step 2: Explain that this allows lifting heavier objects with smaller effort. 3. Step 3: Provide real-life applications such as elevators and cranes. [Ch 7: Work, Energy, and Simple Machines]	
30	What is differentiation in plants, and what changes do cells undergo during this process? Answer: Differentiation is the process by which meristematic tissue becomes specialized to perform specific functions, transforming into permanent tissues. As cells lose their ability to divide, they undergo significant changes. These newly formed cells undergo changes in both structure and function. They become specialized to carry out specific roles such as providing support, transporting water and nutrients, or storing food. For instance, a meristematic cell might develop thick lignin walls to become a hard, dead sclerenchyma cell for strength, or it might form tubular structures to conduct water, thus becoming a permanent part of the plant. ► Why? Differentiation is the structural and functional transformation of meristematic cells into permanent tissues, allowing them to perform specific roles like support, transport, or storage. ► Step-by-step: 1. Step 1: Define differentiation as the process where meristematic tissue becomes specialized permanent tissue. 2. Step 2: Explain that cells lose the ability to divide and undergo structural and functional changes. 3. Step 3: Mention that they become specialized for specific functions like support, transport, or storage. [Ch 3: Tissues in Action]	3
Section D Attempt 4 out of 6.		



Q.NO.	QUESTIONS	MARKS
31	Explain the force of friction and how it affects the motion of objects. When is an object able to overcome friction and start moving? A table is moved across the floor at a constant velocity using a horizontal force F. What is the magnitude of the frictional force exerted by the floor on the table? Justify your answer using Newton's laws. Answer: The force of friction acts on an object in the direction opposite to its direction of motion, or opposite to the direction of the applied force trying to move it. Friction arises between surfaces in contact and depends on the nature of the surfaces in contact. When an object is at rest on the floor and a force is applied on it in the forward direction, the force of friction acts in the opposite direction. The object does not move if the applied force is not large enough to overcome friction. The object starts moving only when the applied force is of larger magnitude than the force of friction, so that a net force acts on the object in the direction of its motion. This is because the motion of an object depends only on the net force acting on it. In addition to the applied force and friction, other forces also act on an object on a surface. The gravitational force (weight) acts in the downward direction, while the surface exerts a normal force in the upward direction perpendicular to the surface. These two forces are balanced and do not affect the horizontal motion. When a table is moved across the floor at a constant velocity using a horizontal force F, the frictional force exerted by the floor on the table is equal to F. By Newton's first law, an object moving with constant velocity has no net force acting on it. Since the table moves at constant velocity, the net horizontal force must be zero, meaning the frictional force exactly balances the applied horizontal force F. ► Why? Friction opposes motion and must be overcome by a larger applied force for an object to start moving. At constant velocity, net force is zero, so friction equals the applied force. ► Step-by-step: 1. Step 1: Define friction and state that it acts opposite to the direction of motion or applied force. 2. Step 2: Explain when an object overcomes friction — applied force must exceed friction to produce a net force. 3. Step 3: Describe the other forces (gravitational force and normal force) and state they are balanced. 4. Step 4: Apply Newton's first law to the table at constant velocity — net force is zero, so friction = applied force F. 5. Step 5: Conclude by linking the result to the concept that motion depends only on net force. [Ch 6: How Forces Affect Motion]	5
	OR Explain the mass by mass percentage and mass by volume percentage methods of expressing the concentration of a solution. A solution contains 5 g of glucose dissolved in water to make 100 mL of solution. Calculate its concentration using the appropriate method and justify why that method is suitable.	



Q.NO.	QUESTIONS	MARKS
	Answer: The concentration of a solution can be expressed in terms of percentage in two main ways: mass by mass percentage and mass by volume percentage, each suited to different practical situations. Mass by mass percentage (% m/m) tells us how many grams of solute are present in 100 grams of the total solution. Mathematically, Mass by mass percentage = (Mass of solute)/(Mass of solution) × 100. This method is commonly used for homogeneous mixtures as well as heterogeneous mixtures such as milk powder and spice mixtures. It is also used to label the composition of packaged foods, showing how much salt, sugar, or protein is present in them. Mass by volume percentage (% m/v) tells us how many grams of solute are present in 100 millilitres of the solution. Mathematically, Mass by volume percentage = (Mass of solute)/(Volume of solution) × 100. This method is used where measuring the volume of a liquid is easier than weighing it, for example, in medicines and laboratories. Calculation: Mass of glucose (solute) = 5 g, Volume of solution = 100 mL. Mass by volume percentage = (5 g)/(100 mL) × 100 = 5% m/v The mass by volume percentage method is appropriate here because glucose solution is a liquid, and measuring its volume is easier than weighing it. A 5% glucose solution is a common example of this method used in medicines and laboratories. Thus, selecting the right method depends on the nature of the solution and the ease of measurement in practical situations. ► <i>Why? The two percentage methods differ in whether the solution's total quantity is measured by mass or volume; mass by volume is preferred for liquid solutions where volume measurement is more convenient.</i> ► Step-by-step: <i>1. Step 1: Define mass by mass percentage and write its mathematical formula.</i> <i>2. Step 2: Explain where mass by mass percentage is used, giving examples like milk powder, spice mixtures, and packaged foods.</i> <i>3. Step 3: Define mass by volume percentage and write its mathematical formula.</i> <i>4. Step 4: Explain where mass by volume percentage is used, giving examples like medicines and laboratories.</i> <i>5. Step 5: Calculate the concentration of the glucose solution using the mass by volume percentage formula.</i> <i>6. Step 6: Justify the choice of method based on the nature of the solution and conclude.</i> <i>[Ch 5: Exploring Mixtures and their Separation]</i>	
32	Explain how a microscope enables us to study cell structure. Describe the three main features that scientists have improved in microscopes over the years, and demonstrate how the estimated size of a cell and its total magnification are calculated.	5



Q.NO.	QUESTIONS	MARKS
	Answer: A microscope allows us to see very small structures clearly and is an essential tool for studying cell structure. Over the years, scientists have turned the microscope into a powerful tool by improving its three main features. Resolution is a measure of clarity of the image, allowing us to distinguish closely spaced structures within a cell. Contrast refers to the difference in brightness between various parts of an object, helping to highlight different cellular components. Magnification is the ability to enlarge the image of a specimen, enabling us to observe details invisible to the naked eye. The estimated size of a cell can be determined using the formula: Estimated size of cell = Diameter of the visible field in micrometre / Number of cells along the diameter. For example, if 25 cells are seen along the diameter of a field of 5000 μm, the size of one cell = $5000 \mu\text{m} / 25 = 200 \mu\text{m}$. The total magnification of a microscope depends on the magnifying power of the eyepiece and the objective lens. If both the eyepiece and the objective lens have a magnifying power of 10X, then the total magnification = $10X \times 10X = 100X$. This means a cell with an estimated size of $200 \mu\text{m}$ will appear 100 times larger. These improvements in resolution, contrast, and magnification have collectively made the microscope a powerful and indispensable tool for studying cells. ► Why? <i>The microscope is essential for cell study because it enhances resolution, contrast, and magnification, and cell size can be quantified using the field diameter divided by the number of cells, while total magnification is the product of eyepiece and objective lens powers.</i> ► Step-by-step: 1. Step 1: State that a microscope is an essential tool for studying cell structure by making very small structures clearly visible. 2. Step 2: Describe the three improved features — resolution (clarity), contrast (brightness difference between parts), and magnification (enlargement of image). 3. Step 3: State the formula for estimated cell size and work through the example ($5000 \mu\text{m} / 25 \text{ cells} = 200 \mu\text{m}$). 4. Step 4: Explain total magnification as the product of eyepiece and objective lens magnifying powers ($10X \times 10X = 100X$). 5. Step 5: Conclude by summarising how improvements in all three features make microscopes powerful tools for cell study. <i>[Ch 2: Cell: The Building Block of Life]</i>	
	OR Describe the step-by-step procedure for plotting a position-time graph from given data. In your explanation, include how to draw the axes, choose suitable scales, plot the data points, and connect them to form the graph.	



Q.NO.	QUESTIONS	MARKS
	Answer: To plot a position-time graph, the first step is to take a sheet of graph paper, which is pre-divided into small squares, making it easier to plot data accurately. On this paper, draw two lines perpendicular to each other. Their point of intersection is known as the origin O. Mark the horizontal line as OX (x-axis) and the vertical line as OY (y-axis). Next, decide which quantity to show along each axis. For motion data, time is shown along the x-axis and position along the y-axis. Then, determine a suitable scale for each quantity so that the data is represented effectively, conveniently, and utilises the available space. For example, the scale could be x-axis: 5 divisions = 1 s and y-axis: 5 divisions = 20 m. Use the chosen scale to mark values for time along the x-axis and for position along the y-axis from the origin. Now, begin plotting points for each set of time and position values. For example, at time 0 s, if position is 0 m, the point is the origin itself. At 1 s and 20 m, draw a line parallel to the y-axis at 1 s on the x-axis and a line parallel to the x-axis from 20 m on the y-axis; their intersection is the required point. Similarly, plot all points corresponding to positions at different instants of time. Once all points are plotted, connect them to create the position-time graph. If the object moves with constant speed, the result is a straight line; if it speeds up, the points form a curve. ► Why? Plotting a graph involves drawing perpendicular axes, choosing scales, marking values, plotting each data point by finding intersections of parallel lines from both axes, and then connecting the points. ► Step-by-step: 1. Step 1: Describe setting up graph paper and drawing perpendicular axes (x-axis and y-axis) with origin O. 2. Step 2: Explain assigning time to x-axis and position to y-axis. 3. Step 3: Explain choosing suitable scales for both axes with an example. 4. Step 4: Describe how to plot each point using lines parallel to the axes and finding their intersection. 5. Step 5: Explain connecting all plotted points to complete the graph and note the shape obtained. [Ch 4: Describing Motion Around Us]	
33	Explain how lateral meristem and intercalary meristem contribute to growth in plants. Describe the location and function of each type, and give examples of everyday observations that demonstrate the action of intercalary meristem.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Meristematic tissues are actively dividing cells responsible for plant growth. Two important types — lateral meristem and intercalary meristem — contribute to growth in different ways. The lateral meristem consists of actively dividing cells arranged in a ring in the stem. These cells divide and produce new cells inside and outside in a concentric manner, which leads to an increase in the diameter (girth) of the stem. This is how plants grow thicker over time. The intercalary meristem is located at the base of the internode or just above the node. A node is the point on a plant stem where branches or leaves arise, and the part of the stem between two nodes is called the internode. When the tip of a young stem is cut, the stem stops growing in length, but new branches arise from the nodes of the stem due to the activity of intercalary meristem. Everyday observations demonstrate the action of intercalary meristem clearly. When a hedge around a garden is cut, after some time more branches appear again, giving the hedge a bushy appearance. Similarly, grass appears again after being mowed or grazed by animals. In both cases, regrowth occurs because of the presence of intercalary meristem at the nodes of the stem. Thus, while lateral meristem is responsible for increase in girth of the stem, intercalary meristem enables regrowth and branching after the plant is cut or grazed, ensuring continued growth and survival. ► Why? Lateral meristem increases stem girth through concentric cell division in a ring, while intercalary meristem at nodes enables regrowth and branching after cutting, as seen in hedges and grass. ► Step-by-step: 1. Step 1: Define lateral meristem — its location (ring in stem) and how it causes increase in girth through concentric cell division. 2. Step 2: Define intercalary meristem — its location (base of internode or just above the node) and its role in regrowth after cutting. 3. Step 3: Explain the terms 'node' (point where branches/leaves arise) and 'internode' (part between two nodes). 4. Step 4: Give examples from the source — hedge becoming bushy after cutting, and grass regrowing after mowing or grazing. 5. Step 5: Conclude by contrasting the two meristems — lateral for girth increase, intercalary for regrowth and branching. [Ch 3: Tissues in Action]	
34	Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess.	5



Q.NO.	QUESTIONS	MARKS
	Answer: In science, laws, theories, and principles are terms with specific and distinct meanings that help organise our understanding of the world in a systematic way. A law usually describes a regular pattern observed in nature, often expressed using words or mathematical relationships. For example, Newton's laws of motion explain the jerk felt when a bus stops suddenly. Laws describe what happens based on repeated observations. A theory goes a step further than a law. It provides an explanation of why those patterns occur, usually based on evidence gathered over time and available at that time. For example, the atomic theory explains how molecules are formed. In science, a theory does not mean a guess or an untested idea; it is an explanation based on careful testing and critical examination. Principles are broad ideas that help us make sense of a given situation. For example, the principle of conservation of energy can be applied when climbing up the stairs. Principles serve as guiding ideas rather than descriptions or explanations. A key feature that makes science reliable is that these ideas are always open to improvement and often change as new evidence becomes available. No scientific theory is ever final and none is beyond question. This openness to being corrected by nature itself is what allows science to help us understand the world we live in. ► Why? Laws describe observed patterns, theories explain why patterns occur based on evidence, and principles are broad guiding ideas. A scientific theory is a well-tested explanation, not a guess. ► Step-by-step: 1. Step 1: Define a law and give the Newton's laws of motion example. 2. Step 2: Define a theory and give the atomic theory example; clarify it is not a guess. 3. Step 3: Define a principle and give the conservation of energy example. 4. Step 4: Explain how all these ideas remain open to improvement as new evidence emerges. 5. Step 5: Conclude by stating why this openness makes science reliable. [Ch 1: Exploration: Entering the World of Secondary Science]	
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Cell Theory and Cell Culture The Cell Theory was formulated through the work of several scientists. In 1838, Matthias Schleiden reported that all plants are made up of cells. In 1839, Theodor Schwann found that all animals are also made up of cells. In 1855, Rudolf Virchow stated that new cells are formed only from pre-existing cells. Building on the understanding of cells, scientists developed cell culture — a method to grow plant and animal cells outside the body. Cells are taken from an organism and placed in a nutrient-rich medium. The right temperature, acidic or alkaline conditions, and moisture under sterile conditions must be maintained to keep the culture safe. (i) Which scientist stated that new cells are formed only from pre-existing cells, and in which year? (ii) State any two principles of the classical Cell Theory. (iii) A laboratory technician maintains proper temperature and nutrient supply but forgets to maintain sterile conditions during cell culture. What is likely to happen and why? (iv) An Austrian botanist proposed that any living plant cell can develop into a complete plant if provided with suitable nutrients and favourable conditions. What is this special ability of plant cells called? Answer: (i) Rudolf Virchow stated this in 1855. (ii) Any two: (1) All living organisms are made up of one or more cells. (2) The cell is the basic unit of structure and function in living beings. (3) All cells arise from pre-existing cells. (iii) The cell culture is likely to get contaminated and the cells may not survive or grow properly. Sterile conditions must be maintained to keep the culture safe; without sterility, contamination can destroy the culture. (iv) Totipotency. ► Why? i) Rudolf Virchow stated this in 1855. ii) Any two: (1) All living organisms are made up of one or more cells. (2) The cell is the basic unit of structure and function in living beings. (3) All cells arise from pre-existing cells. iii) The cell culture is likely to get contaminated and the cells may not survive or grow properly. Sterile conditions must be maintained to keep the culture safe; without sterility, contamination can destroy the culture. iv) Totipotency. [Ch 2: Cell: The Building Block of Life]	5



Q.NO.	QUESTIONS	MARKS
	OR Case Study: Plant Protective and Supporting Tissues A botanist comparing plants from different habitats observes that desert plants have a thick waxy coating on their surface, while aquatic plants float easily on water. She notices that young plant stems bend without breaking, but the hard shells of walnuts are extremely tough and cannot bend. She examines the transverse section of a sunflower stem under a microscope and identifies several types of permanent tissues. She finds that permanent tissues can be simple (composed of only one type of cell) or complex (composed of more than one type of cell), and each tissue is specialised to perform a specific function such as support, transport, or storage. (i) What is the waxy layer of cutin on the epidermis called? (ii) Which type of simple permanent tissue allows plant stems and tendrils to bend without breaking? Name the chemical responsible for this flexibility. (iii) A walnut shell is hard and strong. Identify the tissue, the chemical deposited in its cell walls, and state whether its cells are living or dead. (iv) How does the specialised parenchyma in aquatic plants help them survive in their habitat? Answer: (i) Cuticle (ii) Collenchyma; pectin (a chemical that gives flexibility like rubber) is deposited at the unevenly thickened corners. (iii) Sclerenchyma; lignin is deposited in its cell walls; most of these cells are dead. (iv) Specialised parenchyma forms air spaces which help aquatic plants float. ► Why? i) Cuticle ii) Collenchyma; pectin (a chemical that gives flexibility like rubber) is deposited at the unevenly thickened corners. iii) Sclerenchyma; lignin is deposited in its cell walls; most of these cells are dead. iv) Specialised parenchyma forms air spaces which help aquatic plants float. [Ch 3: Tissues in Action]	
36	Case Study: Kinematic Equations for Vehicles on a Straight Road A motorbike is moving on a straight road with an initial velocity of 28 m/s. The rider applies brakes and the motorbike decelerates uniformly, coming to rest after travelling a distance of 98 m. On a parallel track, a car starts from rest and accelerates uniformly, reaching a final velocity of 24 m/s in 6 seconds. A physics student watching both vehicles decides to compare their accelerations and the distances they cover using the kinematic equations for motion in a straight line with constant acceleration. (i) What is the average acceleration of the car during the 6 seconds? (ii) Using the equation $v^2 = u^2 + 2as$, calculate the acceleration of the motorbike. (iii) Calculate the distance travelled by the car in 6 seconds and the time taken by the motorbike to come to a stop. (iv) Compare the magnitudes of acceleration of the motorbike and the car. Which one is greater?	5



Q.NO.	QUESTIONS	MARKS
	Answer: (i) 4 m/s^2 ($a = (v - u)/t = (24 - 0)/6 = 4 \text{ m/s}^2$) (ii) -4 m/s^2 ($0^2 = 28^2 + 2a(98) \rightarrow 0 = 784 + 196a \rightarrow a = -4 \text{ m/s}^2$. The negative sign indicates deceleration; magnitude is 4 m/s^2.) (iii) For the car: $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(4)(6^2) = 72 \text{ m}$. For the motorbike: $v = u + at \rightarrow 0 = 28 + (-4)t \rightarrow t = 7 \text{ s}$. (iv) Both have the same magnitude of acceleration (4 m/s^2). Neither is greater. ► Why? i) 4 m/s^2 ($a = (v - u)/t = (24 - 0)/6 = 4 \text{ m/s}^2$) ii) -4 m/s^2 ($0^2 = 28^2 + 2a(98) \rightarrow 0 = 784 + 196a \rightarrow a = -4 \text{ m/s}^2$. The negative sign indicates deceleration; magnitude is 4 m/s^2.) iii) For the car: $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(4)(6^2) = 72 \text{ m}$. For the motorbike: $v = u + at \rightarrow 0 = 28 + (-4)t \rightarrow t = 7 \text{ s}$. iv) Both have the same magnitude of acceleration (4 m/s^2). Neither is greater. [Ch 4: Describing Motion Around Us]	
37	Case Study: Scientific Predictions and Testable Questions Rohan predicts it will rain because the clouds look dark. His friend Priya asks him to make his prediction scientifically testable. In science, predictions are not guesses—they are reasoned expectations based on evidence and careful thinking. Good scientific questions look for measurable evidence and past patterns, going beyond simple observations like 'clouds look dark.' When predictions match observations, confidence in the underlying science grows. But when they do not, scientists re-examine their assumptions, models, or measurements. Even weather forecasts can go wrong because tiny differences in conditions can grow over time, yet such failures are not a weakness of science—they are its greatest strength. (i) What is the key difference between a scientific prediction and a guess? (ii) Why are questions about measurable data and past patterns more useful for testing a prediction than simple yes/no questions? (iii) If a scientist's prediction about the weather turns out to be wrong, what steps should the scientist take, and why is this process important for science? (iv) Weather forecasts sometimes fail even though they use measurements and models. Why is this NOT considered a weakness of science? Answer: (i) A scientific prediction is a reasoned expectation based on evidence and careful thinking, whereas a guess is not based on such evidence. (ii) Questions about measurable data and past patterns provide evidence that can be checked and compared, going beyond mere opinion, whereas simple yes/no answers are usually not so useful. (iii) The scientist should re-examine their assumptions, models, or measurements. This is important because when predictions do not match observations, it drives further exploration and deeper understanding, making science reliable and open to correction by evidence rather than opinion. (iv) Because very tiny differences in conditions can grow over time and lead to different outcomes. Such failures are science's greatest strength—scientists do not reject ideas based on opinion but only on evidence, and this openness to being corrected allows science to progress.	5



Q.NO.	QUESTIONS	MARKS
	► Why? i) A scientific prediction is a reasoned expectation based on evidence and careful thinking, whereas a guess is not based on such evidence. ii) Questions about measurable data and past patterns provide evidence that can be checked and compared, going beyond mere opinion, whereas simple yes/no answers are usually not so useful. iii) The scientist should re-examine their assumptions, models, or measurements. This is important because when predictions do not match observations, it drives further exploration and deeper understanding, making science reliable and open to correction by evidence rather than opinion. iv) Because very tiny differences in conditions can grow over time and lead to different outcomes. Such failures are science's greatest strength—scientists do not reject ideas based on opinion but only on evidence, and this openness to being corrected allows science to progress. [Ch 1: Exploration: Entering the World of Secondary Science]	

